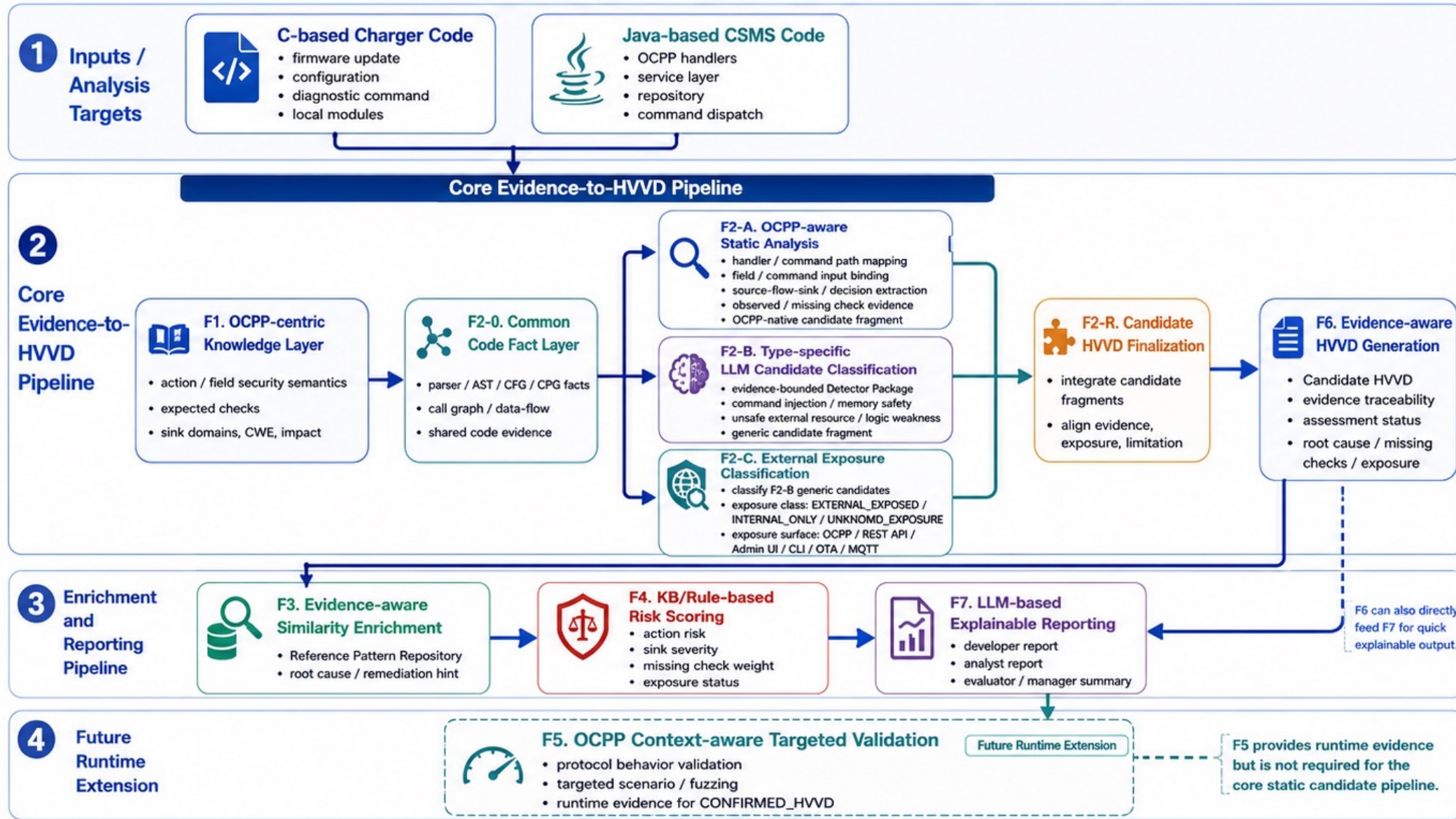


ChargeGuard-HVVD Architecture

Core Architecture

OCPP-aware source-code analysis, type-specific LLM candidate classification, external exposure separation, and evidence-aware HVVD generation



Key Principles



LLM is not a vulnerability confirmer



exposure classification is separated from candidate classification



all outputs are evidence-traceable



CONFIRMED_HVVD requires runtime evidence



Research focus: F1 + F2-A + F2-B + F2-C + F6 | Enrichment: F3/F4/F7 | Optional runtime extension: F5



F2-A. OCPP-aware Static Analysis

OCPP 도메인 지식과 정적 분석 Evidence를 결합한 OCPP-native Candidate 추출 계층

입력 (Inputs)

1 Source Code Repository

- C / C++ / Java
- Charge Point / CSMS

2 F1. OCPP-centric Knowledge Layer

- Action / Field semantics
- Expected checks
- Dangerous sink domains

3 F2-0. Common Code Fact Layer

- Call graph
- Data-flow / AST / CFG
- Function / variable facts

4 Static Analysis Configuration

- Mapping rules
- Pattern rules
- Evidence schema

Core Pipeline

- 1 OCPP Handler / Command Path Mapping**
Inbound handler / Outbound command path
- 2 Payload Field / Command Input Binding**
- 3 Field Semantic Binding**
예: UpdateFirmware.location → firmware_download_url
- 4 Source-Flow-Sink / Source-Decision Extraction**
- 5 Dangerous Sink / Security-sensitive Decision Mapping**
system(), URL.openConnection(), authorization decision, state transition
- 6 Observed Check Detection**
 - Rule-based Check Pattern Detection
 - Optional LLM-assisted Check Semantics Classification
- 7 Expected Check Matching**
- 8 Missing / Weak / Unverified Check Candidate Generation**
- 9 OCPP-native Evidence Package**
- 10 OCPP-native Candidate HVVD Fragment**

LLM은 취약점 탐지기가 아니라 Validation Checker로 제한

Primary Outputs

- ocpp_source_map.json
- handler_map.json
- field_binding_map.json
- ocpp_flow_candidates.jsonl
- observed_check_analysis.jsonl
- expected_check_matching_results.jsonl
- missing_check_candidates.jsonl
- ocpp_native_candidate_fragments.jsonl
- ocpp_evidence_packages.jsonl

★ 핵심 설계 원칙

1 OCPP action/field에서 출발하는 OCPP-native evidence extraction

2 Inbound / Outbound OCPP path 모두 지원

3 취약점 확정기가 아닌 Evidence Extraction Layer

4 Rule-based + Optional LLM-assisted Validation Checker

5 F6 Evidence-aware HVVD Generation의 입력으로 활용

→ F6. Evidence-aware HVVD Generation

F2-B Type-Specific LLM Vulnerability Candidate Classification

ChargeGuard-HVVD

General-purpose LLM engine controlled by detector packages, static evidence, and structured output rules

